

Banking Analytics Dashboard

Problem Statement:

Develop a basic understanding of risk analytics in banking and financial services and understand how data is used to minimize the risk of losing money while lending to customers.

Solution:

With our dashboards which are created using Power BI latest tools helps the company to make a decision based on the applicant's profile like if the applicant is likely to repay the loan, then approving the loan otherwise not.

About Dataset:

This dataset basically contains information about bank details, various client details which consists of multiple tables which are interlinked with each other through keys like primary key and foreign key.

The various tables are Banking Relationship, Client-Banking, Gender, Investment Advisor and Period.

Project Objectives:

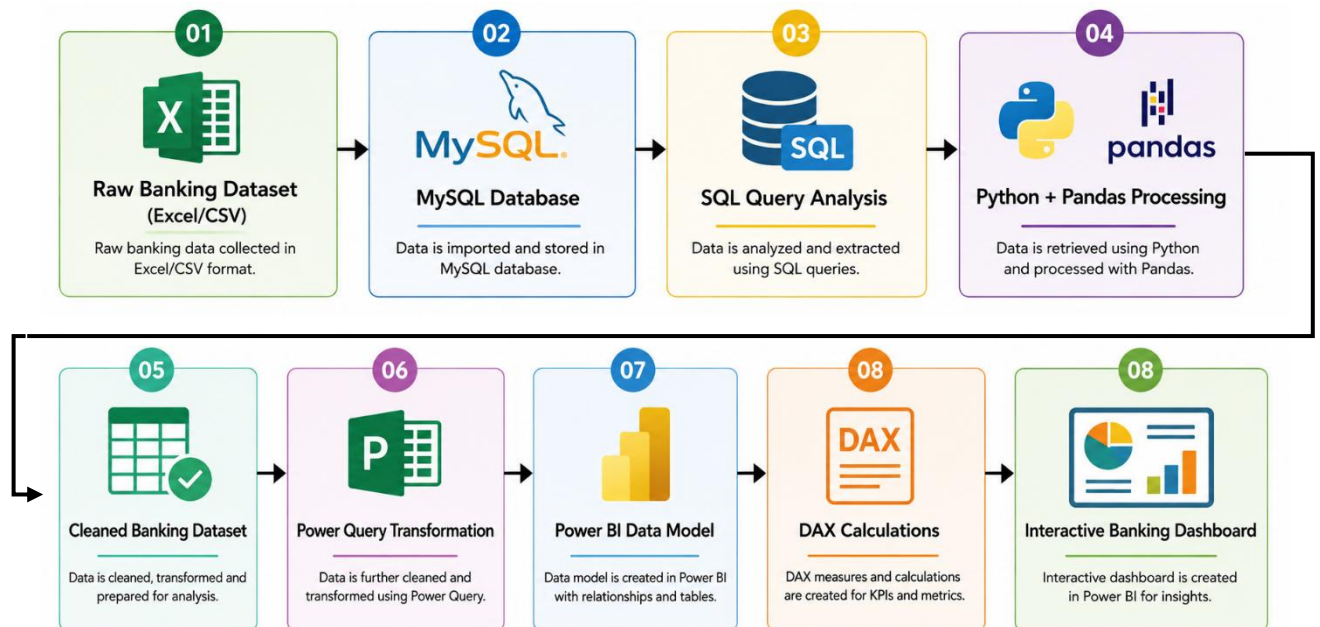
1. **To analyze banking data** by collecting, processing, and visualizing customer, loan, and deposit information for meaningful business insights.
2. **To develop an interactive Power BI dashboard** that enables users to monitor key banking performance indicators (KPIs) and make data-driven decisions.
3. **To preprocess and transform banking data** using Python and MySQL, ensuring data accuracy, consistency, and reliability before analysis.
4. **To create dynamic visualizations and DAX measures** for analyzing loan performance, deposit trends, customer engagement, and business lending.
5. **To provide actionable insights** that help banking professionals identify trends, monitor financial performance, and improve strategic decision-making.

Tools Used:

Tool	Work Done
Microsoft Excel (CSV Dataset)	Data Source - Stores the raw banking customer and transaction data before importing into MySQL.
MySQL	Database Management - Stores structured banking data, creates tables, performs SQL queries, and manages relationships. Database Development - Used to create databases, import datasets, execute SQL queries, and validate stored data. <u>SQL:</u> Data Analysis - Retrieves, filters, aggregates, and summarizes banking information using queries. <u>MySQL Connector:</u> Database Connectivity - Establishes communication between Python and MySQL to fetch and load data.

Python	Data Processing - Connects to MySQL, extracts data, performs preprocessing, and prepares datasets for visualization. Jupyter Notebook: Python Development - Executes Python scripts, connects to databases, performs analysis, and exports processed data. Pandas: Data Manipulation - Cleans datasets, handles missing values, transforms columns, and performs aggregations. NumPy: Numerical Computing - Supports mathematical and statistical calculations on banking data.
Power BI Desktop	Dashboard Development - Creates interactive dashboards and business intelligence reports using processed banking data. Power Query: ETL (Extract, Transform, Load) - Cleans, transforms, and prepares imported data before visualization. DAX (Data Analysis Expressions): Business Calculations - Creates KPIs, calculated columns, calculated tables, and dynamic measures.

Block Diagram:



Procedure:

1. Excel:

Microsoft Excel was used as the primary data source for the banking project. The following tasks were performed:

- Collected and stored the raw banking dataset.
- Organized customer, loan, and deposit records into a structured format.
- Verified data for missing values, duplicates, and inconsistencies.

- Performed basic formatting and data validation.
- Saved the cleaned dataset in **CSV/Excel** format for importing into **MySQL** and further processing.
This provided a well-structured and reliable dataset for subsequent analysis using MySQL, Python, and Power BI.

2. **MySQL:**

MySQL was used to store and manage the banking dataset in a structured relational database. The following tasks were performed:

- Created the **Banking** database and required tables.
- Imported the banking dataset into MySQL.
- Retrieved data using **SQL queries**.
- Performed filtering, sorting, grouping, and aggregation of records.
- Calculated business metrics such as total loans, total deposits, and customer counts.
- Validated the data before exporting it for Python processing and Power BI visualization.

This ensured efficient data storage, retrieval, and analysis throughout the project.

3. **Python:**

Python was used to process and prepare the banking data before visualization in Power BI. The following tasks were performed:

- Connected to the MySQL database and imported banking data.
- Loaded the data into **Pandas Data Frames** for analysis.
- Cleaned the dataset by removing duplicates and handling missing values.
- Converted data into appropriate formats and standardized columns.
- Performed **Exploratory Data Analysis (EDA)** to understand loan, deposit, and customer trends.
- Created summary statistics and aggregated data for business insights.
- Validated the processed data for accuracy and consistency.
- Exported the cleaned dataset for use in **Power BI**, where interactive dashboards and reports were developed.

This preprocessing ensured that the data was accurate, consistent, and ready for effective business analysis.

4. **Power BI:**

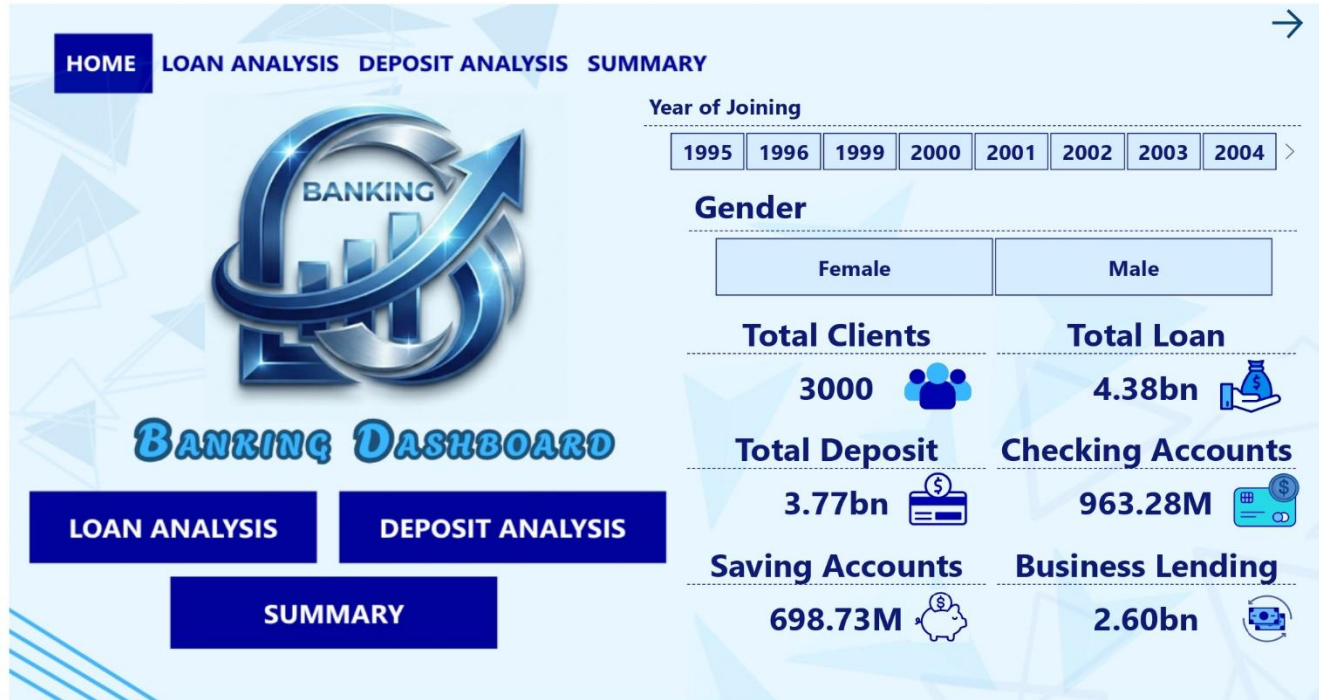
Power BI was used to transform the processed banking data into an interactive and user-friendly dashboard for business analysis. After importing the cleaned dataset, **Power Query** was used to perform additional data transformations, such as changing data types, renaming columns, removing unnecessary fields, and preparing the data for analysis. A **data model** was then created by establishing relationships between the tables to enable accurate reporting.

Using **DAX (Data Analysis Expressions)**, calculated columns and measures were created to generate key performance indicators (KPIs) such as **Total Loans, Total Deposits, Business Lending, Client Count, Average Loan, Average Deposit, and Engagement Days**. These dynamic measures enabled real-time calculations based on user-selected filters.

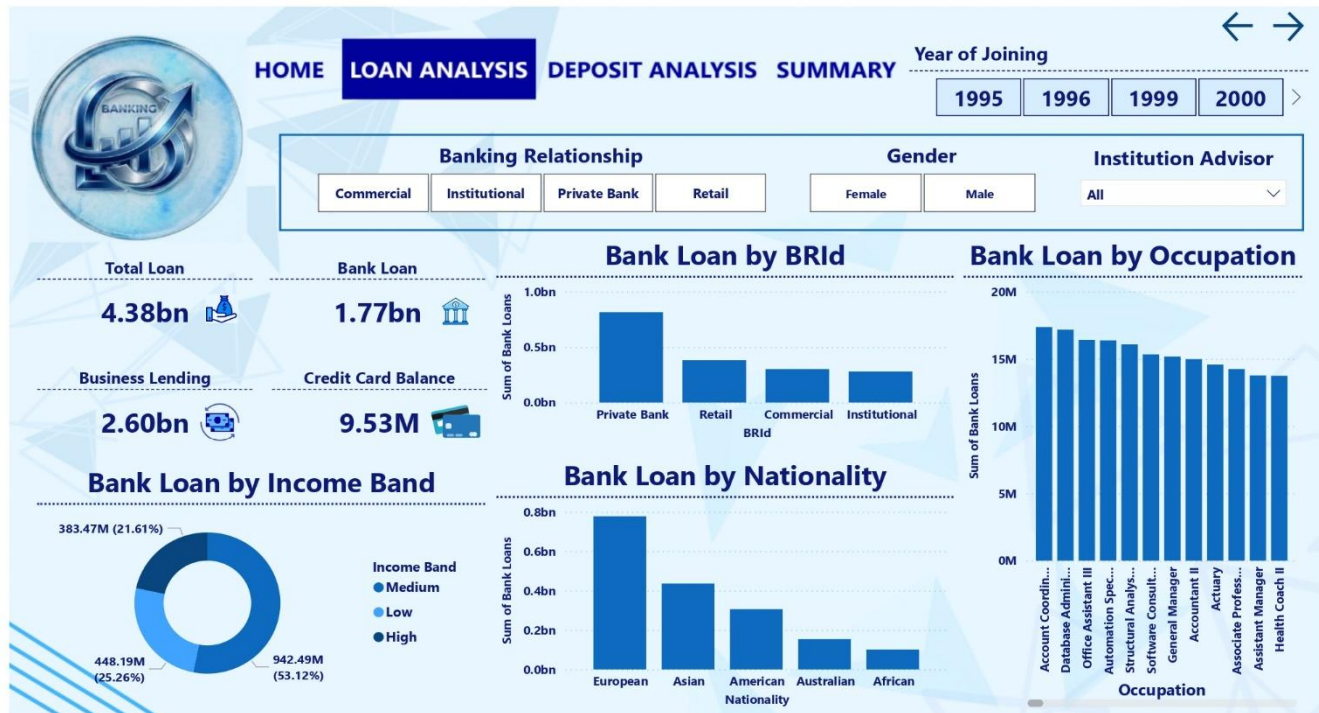
The dashboard was designed with multiple interactive report pages, including **Home, Loan Analysis, Deposit Analysis, and Summary**. Various visualizations such as **KPI cards, clustered column charts, donut charts, bar charts, tables, and slicers** were used to present insights clearly. Interactive filters and navigation buttons were added to allow users to explore data by different customer attributes and banking relationships.

The below given pictures show the dashboard:

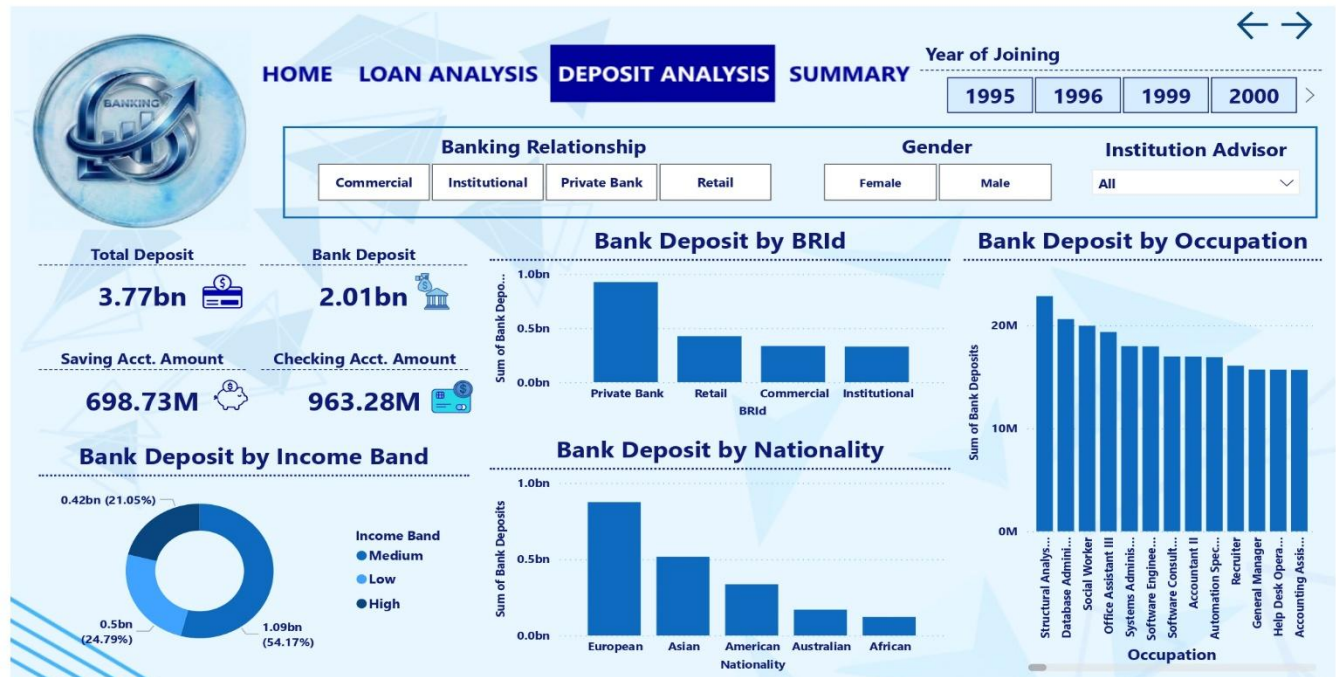
Home Page:



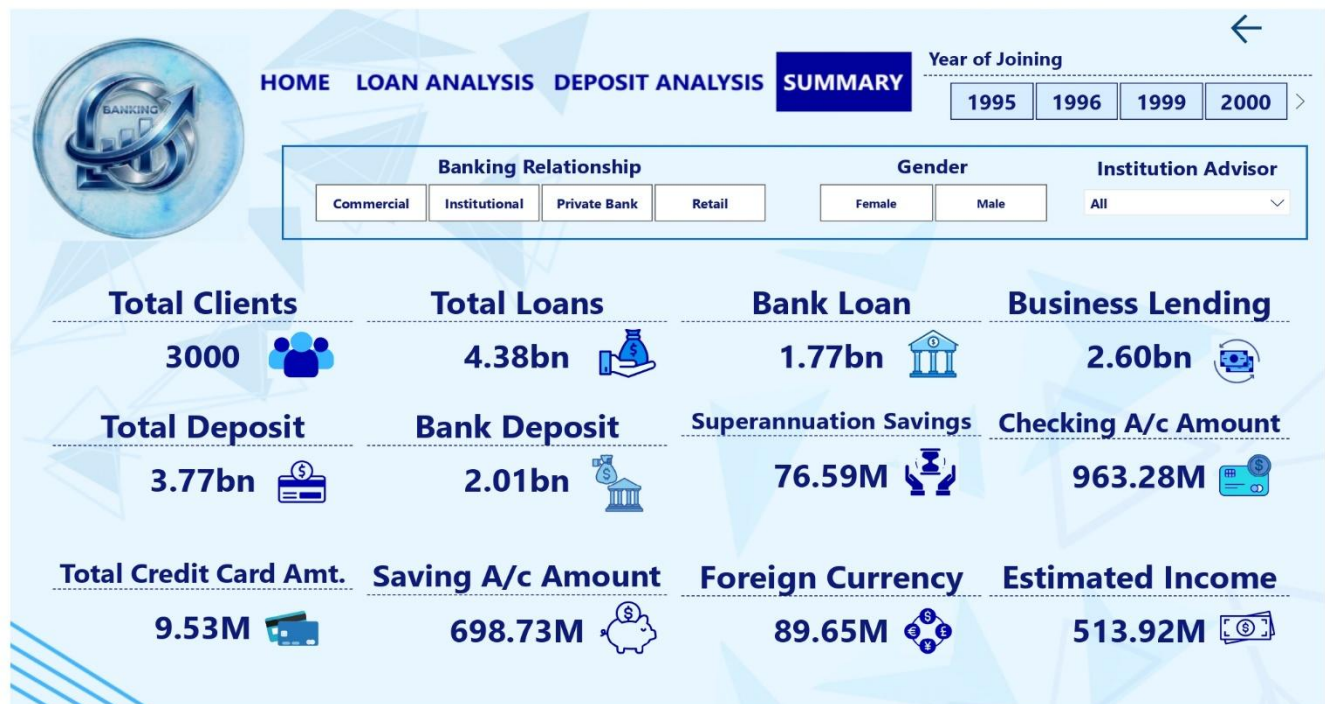
Loan Analysis:



Deposit Analysis:



Summary:



Overall, Power BI converted raw banking data into meaningful visual reports, helping users monitor financial performance, identify trends, analyze customer behavior, and support data-driven decision-making.

Conclusion:

Empowered by the latest data visualization techniques, Power BI dashboards are among the most effective resources for using in banking sector. As outlined in this write-up, a banking operations dashboard in Power BI can be developed with key banking related metrics and KPIs.

Future Work:

With these dashboards banks can easily know what is the total loan amount and all other things of a particular investor. It also helps which type of banks have a greater number of clients as we can see private banks have a greater number of clients so it can help other banks can build their strategies to increase clients. It also provides insights about which nationality has highest bank loans. It gives information about various types of amounts involved in different types of accounts by investors.